Structural Selection at Fixed Energy

Evidence Beyond Energy-Based Ranking in Nonlinear Field Systems

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Abstract

Energy minimisation is one of the central organising principles in classical field theory. However, equal or nearly equal energy does not necessarily imply equal structural quality. This paper presents a compact numerical benchmark for this distinction.

We define an operational structural functional, F_{MAAT}, built from dimensionless support factors for equation consistency, conservation consistency, controlled activity, connectivity, and robustness. The functional is evaluated on static ϕ^4 kink and domain-wall configurations in one and two spatial dimensions, on a fixed-energy ϕ^4 perturbation ensemble, and on a Sine–Gordon soliton benchmark.

The central result is that configurations with approximately identical total energy can receive strongly different structural scores. In particular, smooth localized perturbations and rough noise can be energy-matched while remaining clearly separated by F_{MAAT}. Across robustness sweeps over grid size, random seed, and deformation amplitude, the active structural ordering is stable in 1800/1800 tests.

The results do not establish a fundamental theory of nature. They provide a minimal reproducible test showing that structural selection is not reducible to energy ranking alone.

1 Status and Purpose

This paper is a numerical benchmark note. It is intended to test one precise claim:

> *A structural selection functional can distinguish coherent from rough field configurations even when ordinary energy is approximately fixed.*

The functional used below is operational and phenomenological. It is not a first-principles microscopic derivation, and it is not claimed to replace the standard action principle. The separation used here is:

$$S_0[\Phi] \quad \text{defines the field dynamics, while} \quad F_{\mathrm{MAAT}}[\Phi] \quad \text{ranks structural quality.}$$

Thus F_{MAAT} is a selection diagnostic on a configuration ensemble, not a new fundamental equation of motion.

2 Structural Functional

For each structural sector a, define a non-negative defect $d_a[\Phi]$ and the corresponding bounded support factor

$$\Gamma_a[\Phi] = \frac{1}{1 + d_a[\Phi]}. \tag{1}$$

The operational structural score is

$$\boxed{F_{\mathrm{MAAT}}[\Phi] = -\sum_{a\in\{H,B,S,V,R\}} \lambda_a \log(\epsilon + \Gamma_a[\Phi])} \tag{2}$$

with $\epsilon > 0$ a small numerical floor. Lower F_{MAAT} corresponds to higher structural support.

The five diagnostic sectors are:

- H: equation consistency,
- B: conservation consistency,
- S: controlled activity, implemented here as bounded field variance or nontrivial excitation,
- V: nearest-neighbour structural connectivity,
- R: robustness or smoothness under local perturbations.

For the static scalar-field tests below we use the explicit proxies

$$\begin{aligned} H &= \frac{1}{1+\langle r[\phi]^2\rangle}, & B &= 1, \\ S &= \frac{\mathrm{Var}(\phi)}{1+\mathrm{Var}(\phi)}, & V &= \frac{I_{\mathrm{nn}}}{1+I_{\mathrm{nn}}}, \\ R &= \frac{1}{1+\langle(\Delta\phi)^2\rangle}, \end{aligned} \tag{3}$$

where $r[\phi]$ is the static equation residual and I_{nn} is a nearest-neighbour correlation proxy. In these static scalar benchmarks the conservation sector is intentionally set to $B = 1$, because no independent time-dependent conservation residual is being tested. In dynamical systems, B becomes a nontrivial sector measuring energy–momentum or constraint closure.

The activity sector S should not be read as an independent physical field. It is a bounded excitation proxy: configurations with exactly vanishing variance are structurally inactive, while nontrivial coherent patterns receive finite support without allowing unbounded activity to dominate the score.

The accompanying stability score is

$$\mathrm{Stability}(\Phi) = \min\Big(R,\ (HBSV)^{1/4}\Big)\,. \tag{4}$$

The vacuum is treated as a special inactive reference: it is exactly consistent, but its activity and connectivity supports vanish. The primary ranking test is therefore the ordering among active structures.

3 Field Models

3.1 ϕ^4 Model

The static ϕ^4 energy functional is

$$E_{\phi^4}[\phi] = \int \left[\frac{1}{2}|\nabla\phi|^2 + \frac{1}{8}(\phi^2-1)^2\right] \mathrm{d}^d x. \tag{5}$$

The corresponding static residual used in the H sector is

$$r_{\phi^4}[\phi] = -\Delta\phi + \frac{1}{2}\phi(\phi^2-1). \tag{6}$$

In one dimension the reference kink is $\phi(x) = \tanh(x/2)$. In two dimensions the reference domain wall is $\phi(x,y) = \tanh(x/2)$.

3.2 Sine–Gordon Model

As a cross-model check, the Sine–Gordon energy is

$$E_{\mathrm{SG}}[\phi] = \int \left[\frac{1}{2}\phi_x^2 + 1 - \cos\phi\right] \mathrm{d}x, \tag{7}$$

with residual

$$r_{\mathrm{SG}}[\phi] = -\phi_{xx} + \sin\phi. \tag{8}$$

The reference soliton is $\phi(x) = 4\arctan(e^x)$.

4 Numerical Protocol

All derivatives are evaluated using open-boundary finite differences. The test ensemble consists of:

- a one-dimensional ϕ^4 kink, distorted kink, strongly distorted kink, and chaotic field;
- a two-dimensional ϕ^4 domain wall, distorted wall, strongly distorted wall, and chaotic field;
- a fixed-energy one-dimensional ϕ^4 perturbation ensemble;
- a Sine–Gordon soliton, distorted soliton, strongly distorted soliton, and chaotic field.

For the fixed-energy test, deformation amplitudes are chosen by bisection so that the total energies match a common target energy to numerical precision. The code records all scores in JSON and generates the figures shown below.

5 Results

5.1 ϕ^4 Structural Ranking

In one dimension the active structural ordering is

$$F_{\mathrm{MAAT}}(\text{kink}) < F_{\mathrm{MAAT}}(\text{distorted kink}) < F_{\mathrm{MAAT}}(\text{strong distorted kink}) < F_{\mathrm{MAAT}}(\text{chaos}).$$

In two dimensions the corresponding ordering is

$$\begin{aligned} F_{\mathrm{MAAT}}(\text{domain wall}) &< F_{\mathrm{MAAT}}(\text{distorted wall}) \\ &< F_{\mathrm{MAAT}}(\text{strong distorted wall}) < F_{\mathrm{MAAT}}(\text{chaos}). \end{aligned} \tag{9}$$

State	**Energy**	$\boldsymbol{F_{\mathrm{MAAT}}}$	**Stability**
kink	0.666633	1.510435	0.686636
distorted kink	0.955515	1.982221	0.646850
strong distorted kink	5.454418	6.641044	0.076840
chaos	1051.783483	46.422606	0.000001
domain wall	13.410125	1.509847	0.686730
distorted domain wall	13.597402	1.510410	0.686713
strong distorted domain wall	15.645638	1.531145	0.685270
chaos (2D)	4441.853365	28.903067	0.000037

Table 1: Representative structural rankings in one- and two-dimensional ϕ^4 benchmarks.

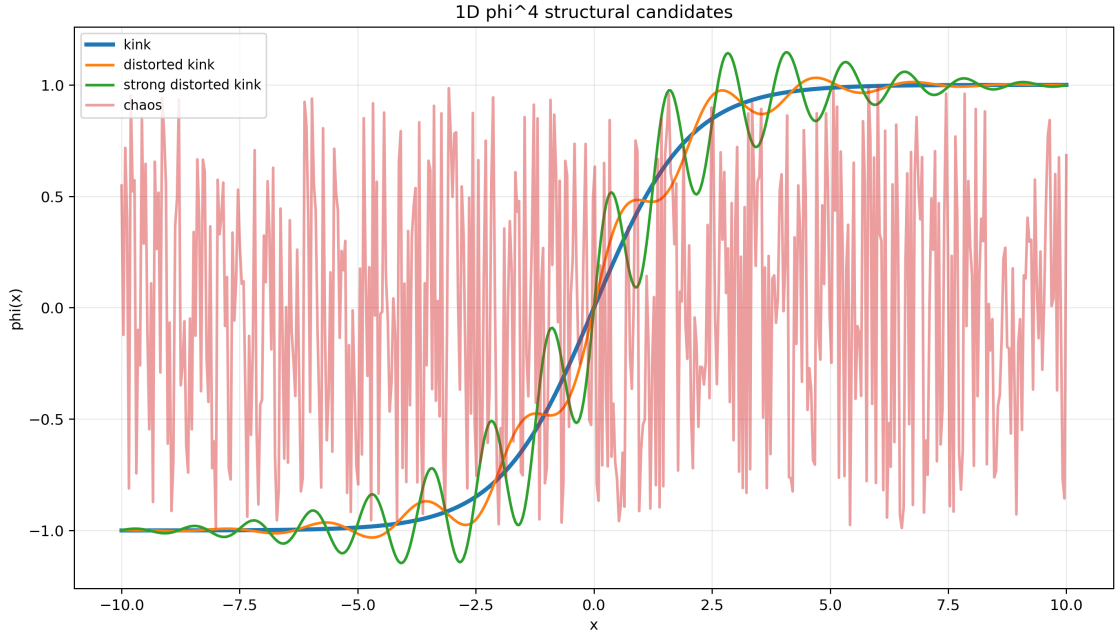

Figure 1: One-dimensional ϕ^4 configurations used in the structural ranking benchmark.

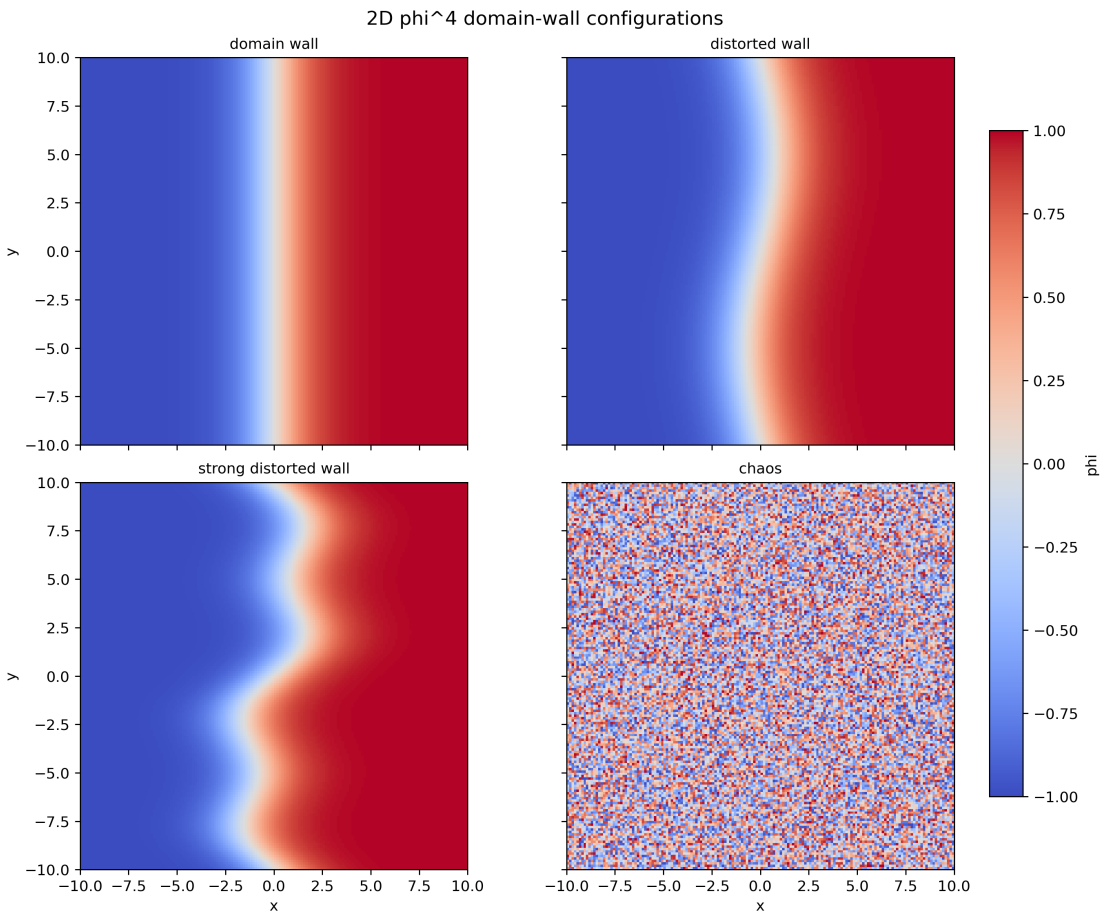

Figure 2: Two-dimensional ϕ^4 domain-wall configurations. The layout uses separated panels and a shared colour bar to avoid the label overlap present in the first draft of the figure.

5.2 Energy Versus Structural Ranking

Energy and structural ranking do not encode the same ordering. The vacuum has minimal energy, but it is structurally inactive because S and V vanish. Conversely, chaotic configurations can carry large activity but are penalised through equation residuals and robustness defects.

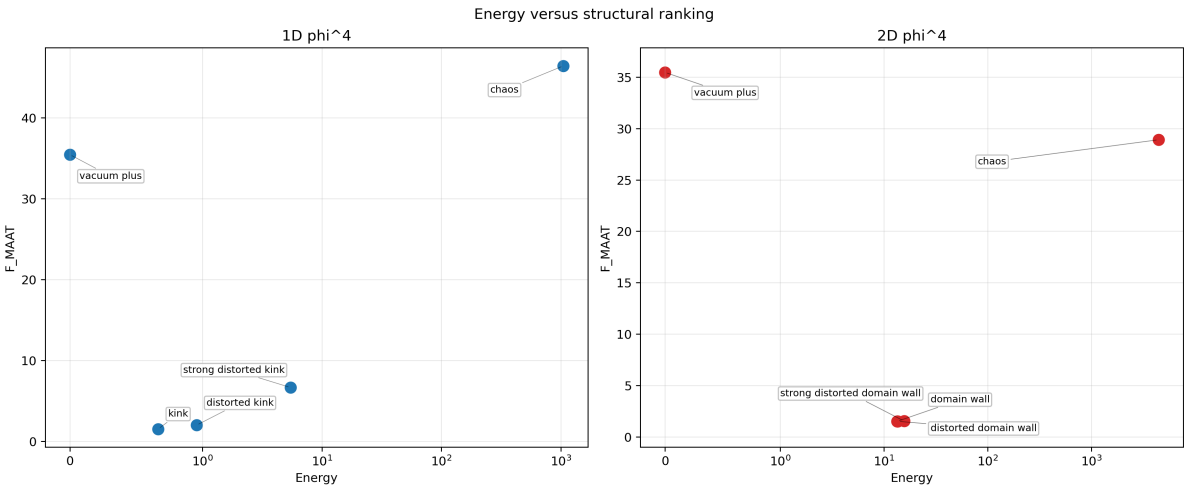

Figure 3: Energy versus F_{MAAT} in the one- and two-dimensional ϕ^4 benchmarks. The separated 2D labels make the near-degenerate domain-wall points visible without overlap.

5.3 Fixed-Energy Test

The central test constructs several one-dimensional ϕ^4 perturbations with nearly identical energy. Their total energies differ only at the 10^{-5} level, while their structural scores differ by almost an order of magnitude.

State	Energy	F_{MAAT}	Stability
two-bump	0.966605	1.539255	0.686698
localized bump	0.966578	1.595588	0.683180
smooth wave	0.966628	1.998289	0.645537
rough noise	0.966626	12.155561	0.004863

Table 2: Fixed-energy comparison. The four perturbations have approximately the same energy, but strongly different structural scores.

5.4 Sine–Gordon Cross-Model Check

The same diagnostic logic also produces the expected active ordering in the Sine–Gordon model:

$$\begin{aligned} F_{\mathrm{MAAT}}(\text{soliton}) &< F_{\mathrm{MAAT}}(\text{distorted soliton}) \\ &< F_{\mathrm{MAAT}}(\text{strong distorted soliton}) < F_{\mathrm{MAAT}}(\text{chaos}). \end{aligned} \tag{10}$$

6 Robustness

Robustness sweeps were performed over grid size, random seed, and deformation amplitude. The active structural ordering passed in all tested cases:

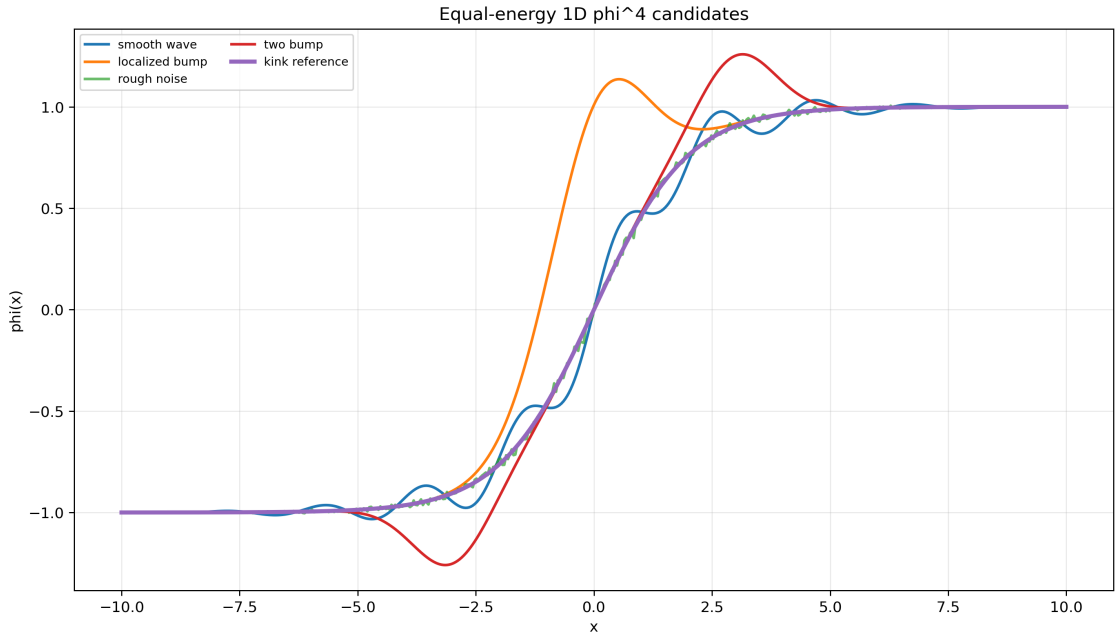

Figure 4: Fixed-energy one-dimensional ϕ^4 perturbations.

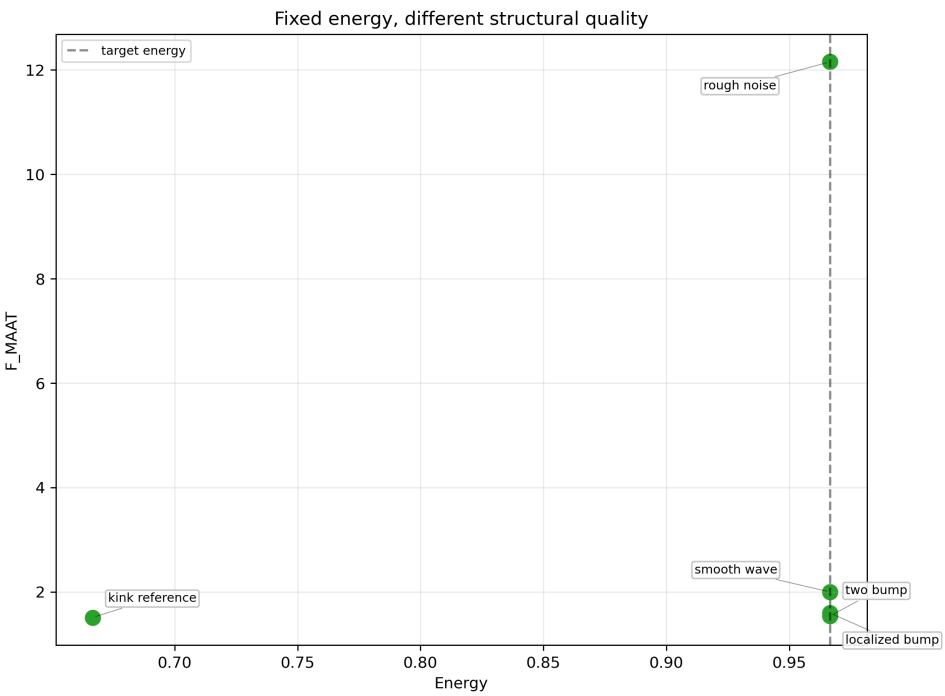

Figure 5: Fixed-energy test: energy is approximately fixed while F_{MAAT} separates coherent perturbations from rough noise.

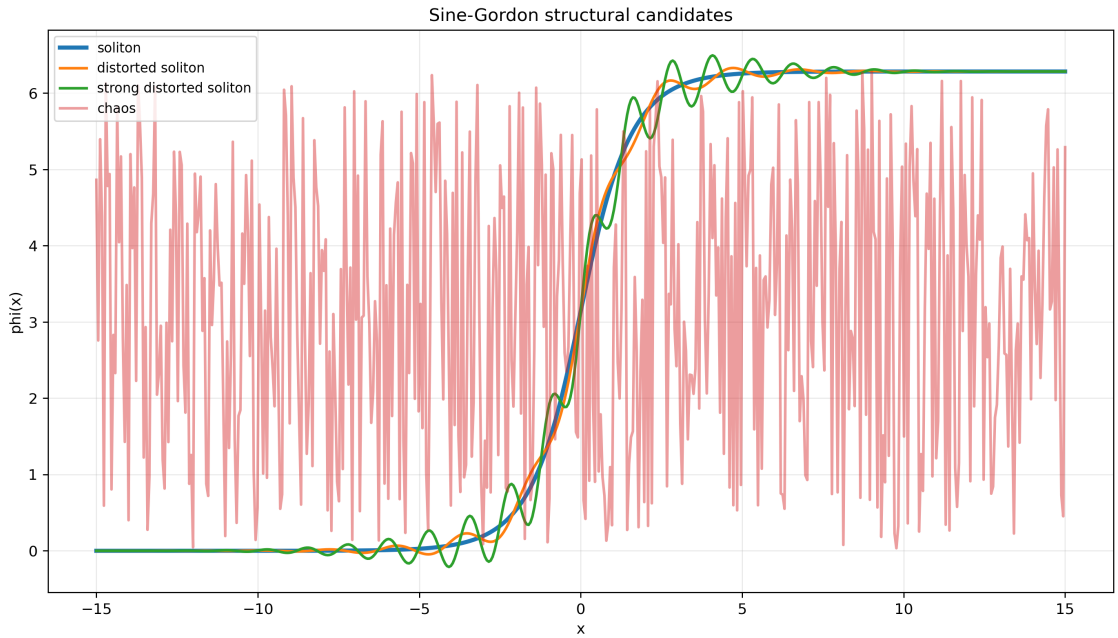

Figure 6: Sine–Gordon soliton configurations used as a cross-model check.

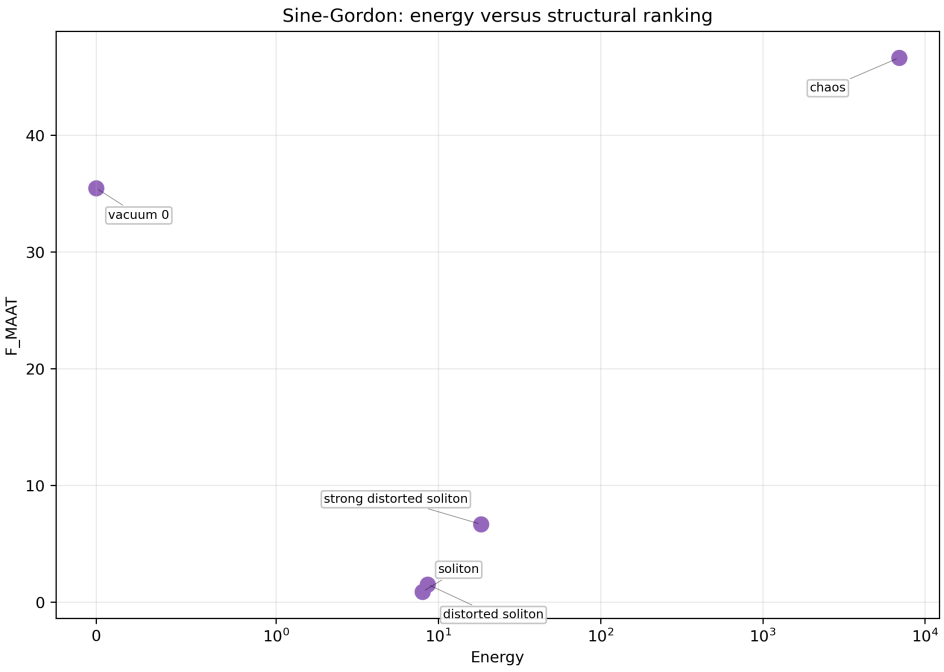

Figure 7: Energy versus structural ranking in the Sine–Gordon benchmark.

Benchmark	Tests	Passed	Pass rate
1D ϕ^4	720	720	100%
2D ϕ^4	360	360	100%
Sine–Gordon	720	720	100%
Combined	1800	1800	100%

This does not prove universality. It shows that the proposed structural ordering is not a fragile artefact of one grid size, one seed, or one deformation amplitude within these toy benchmarks.

7 Interpretation

The fixed-energy result is the key point. If F_{MAAT} were merely a disguised energy functional, configurations with matched energy would receive comparable scores. They do not. Rough noise is strongly penalised because equation consistency and robustness collapse, while smooth localized perturbations retain high structural support.

The physical reason is that the two quantities encode different information. Energy is a global scalar cost obtained by integrating a density over the configuration. The structural score is local and relational: it depends on equation residuals, smoothness, bounded excitation, and nearest-neighbour correlations. These diagnostics can change strongly while total energy is held approximately fixed.

This demonstrates that structural information is not contained in the energy functional alone.

The benchmark therefore supports the following limited conclusion:

> *Energy controls admissibility and cost, but it does not fully determine structural coherence.*

8 Code and Reproducibility

The consolidated script used for the results in this paper is:

`structural_selection_fixed_energy_benchmarks.py`

It generates:

`fixed_energy_structural_selection_results.json`

and all figures in:

`fixed_energy_structural_selection_plots/`.

The repository-level reproducibility entry point is:

`experiments/README.md` in
`https://github.com/Chris4081/structural-selection-principle`.

9 Limitations

The present benchmark has several limitations:

- The diagnostics are operational proxies, not microscopic derivations.
- The tests use simple scalar field models and do not establish a complete theory of nature.
- The conservation sector is trivial in the static scalar benchmarks ($B = 1$); dynamical tests require a nontrivial conservation residual.

- The vacuum is treated as an inactive reference rather than as a competing active structure.
- Experimental validation and first-principles derivation remain open.

10 Conclusion

The numerical evidence presented here shows that structural selection is not reducible to energy minimisation in these benchmarks. In particular, approximately equal-energy field configurations can have sharply different structural scores.

This makes the fixed-energy test a useful validation step for the broader structural-selection programme: it isolates the claim that structural coherence contains information not captured by total energy alone.

References

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